

Stock Market Efficiency and Strategies for Developing GCC Financial Markets: *A Case - study of Bahrain Stock Market*[#]

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Abstract

The paper presents methodology, results of an empirical study for testing the weak form of Efficient Market Hypothesis of Bahrain Stock Market. The paper also reviews and highlights some of the recent strategic initiatives ushered in by the Bahrain Stock Exchange to expand and strengthen Bahrain's financial market in terms of listing procedures, regulatory mechanism, and trading & settlement procedures, which have implications and learning for the stock exchanges of GCC countries.

Further, strategies for developing GCC financial markets have been suggested based upon the experiences of other developing and developed financial markets.

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1 Introduction

The question of market efficiency is of fundamental importance to financial managers due to ‘net present value investment’ criterion, timing of financing decisions and mergers & acquisitions as these assume existence of efficient financial markets.

An efficient stock market is a market in which current securities’ prices reflect all available relevant information and the equity prices will fully reflect the favourable effect of profitable capital investments in fixed assets. Conversely, unwise or unprofitable capital investment decisions will reduce the market value of the firm. In the words of Aswath Damodaran (1997), an efficient market is one in which the market price is an unbiased estimate of the true value of the investment.

It is no exaggeration to say that in a modern industrialist society, which recognizes the rights of private ownership or capital, stock exchanges are not simply a convenience, but they are

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essential. In fact, they are markets, which exist to facilitate purchase and sale of shares, securities or bonds issued by the Government and/or Joint Stock Companies. Thus, stock exchange has a greater role not only in shaping the economy of any nation, but also ensuring an efficient market. As the country moves towards the socialistic pattern of society like the State of Bahrain, this tendency is certain to be strengthened.

In this context, an attempt is made in this paper to present the results of an empirical study to test the weak form of (Efficient Market Hypothesis) Bahrain Stock Market and review the recent initiatives by Bahrain Stock Exchange to develop Bahrain financial market. Besides this, an attempt is also made to delineate appropriate strategies, having implications for developing and integrating financial markets in GCC countries.

2. Research Methodology

In order to find the gap between the past and present studies and to establish the relevance of the present study, an attempt is also made here to review the literature related to the stock market efficiency.

Several studies by Douglas and Vance (1983), Kamarotou and O'Hanlon (1989), Hiroyuki and Matthew (1999), Neil and Robert (1998), Terrence and Mario (1999), Urruita (1995), Lo and MacKinlay (1988), Steven M. Dawson (1984), Fama (1969), have focused the informational efficiency of matured stock markets.

Chung and Tai (1999), Rajen and Qiao (1997), Ling (1997) focused their study on the relationship between market efficiency and factors such as money supply, inflation, interest rates etc.

Dahel & Laabas (1999), Nabeel (1995, 1999), Nabeel and Moosa (1997), Butler and Malaikh (1992) have examined the market efficiency in some of the GCC markets.

Though ample empirical evidence is available to support one or the other form of Efficient Market Hypothesis, no comprehensive study has been made so far particularly in the context of relatively new and emerging stock markets such as Bahrain Stock Market (established in 1989), focusing on market efficiency, regulatory mechanism and reforms to develop financial markets. The present study focusing on Bahrain stock market is an attempt in this direction and is topical as the Kingdom of Bahrain is in the process of emerging as a very strong financial center in the Gulf region.

The study proceeded with setting-up a **null hypothesis** that the *'Bahrain Stock Market is efficient in weak form (current stock prices fully reflect all the information that is contained in the historical sequence of prices and hence there is no benefit in forecasting the future prices).'*

As efficient stock markets reflect the health of the economy of a country, the study besides focusing on testing market efficiency also reviewed the recent financial reforms initiated to further improve market efficiency. Further, an attempt has also been made to propose appropriate strategies for developing and integrating financial markets in GCC countries.

Serial Correlation Tests and Runs Tests have been performed on the two sets of daily share price changes and percentage change in market returns, based on daily prices of individual firms and BSE Daily Index respectively. For this purpose, 12 top companies out of 36 that were listed with BSE in terms of market capitalization, volume and value of shares traded for the year 2000 are chosen, the details are presented in Table - 1. The select 12 companies (sample size) account for 80-90% of the total BSE's transactions during the time frame (1996-2000) chosen for the purpose of this study.

Table 1
Market capitalization, volume and value of shares traded
by the select companies during the year 2000.

Company	Market Capitalization (millions of BD)	Volume (millions of shares)	Value (millions of BD)
Al-Ahli Commercial Bank	64.680	62.219	28.838
Arab insurance Group	82.789	40.429	14.427
Bank of Bahrain & Kuwait	134.299	27.316	6.052
Bahrain Islamic Bank	52.440	20.768	4.973
Bahrain Commercial Facilities	23.600	11.077	2.713
Bahrain Intl. Bank	103.384	68.307	10.367
Bahrain Middle East Bank	62.840	147.899	27.355
Bahrain Telecommunications	615	35.514	21.303
Investcorp Bank	533.832	14.656	20.042
National Bank of Bahrain	168	11.062	4.598
National Import & Export	48.279	8.032	4.960
United Gulf Bank	121.6	18.665	2.330

Source: Market Activity by Company, Statistical Highlights for the year 2000, published by Bahrain Stock Exchange.

The two sets of data, company share prices and BSE index were collected from the Bahrain Stock Exchange, The Gulf Daily News and The Bahrain Tribune. The data has been analyzed using SPSS software package for computing correlation coefficients and MS Excel for computing daily price changes, percentage change in market returns.

The study is constrained by the limitations of time series analysis, statistical bias of a thinly traded stock market. However, sufficient care has been taken to validate data collected from the secondary sources and other limitations encountered in the study are discussed at appropriate places in the paper.

3 Empirical Results and Analysis

3.1 Serial Correlation Tests based on changes in daily share prices

For each of the 12 firms selected, Serial Correlation Tests have been conducted for the daily share price changes over 12 months (starting from January to December) of a particular year (for example, year 1996). The daily share price changes over a month is represented as Var01 to Var60 to indicate the share price changes for the months January, 1996 (Var01) to December, 2000 (Var60).

The flexibility of SPSS package facilitated easy computation of the correlation coefficients among variables (ranging from Var01 to Var60). The variables assigned to the months of each year studied are presented in Table-2.

Table 2
The variables assigned to the months of each study year

Variables assigned	Months & Year
01 – 12	January - December, 1996
13 – 24	January - December 1997
25 – 36	January - December 1998
37 – 48	January - December 1999
49 – 60	January - December 2000

The correlation coefficients among the set of daily share price changes over a month (for example: January, 1996) and the sets of daily share price changes over the remaining months of the year 1996 (February to December) have been computed using SPSS software package. This in effect amounted to computing correlation coefficients over 12 lag periods, each lag period being a month.

The correlation coefficients for the remaining years 1997-2000 were also computed using the same procedure.

The results of Serial correlation Tests (correlation coefficients) over the 5-year period (consisting of 60 lag periods) for all the 12 companies have been produced in the standard format of SPSS output (SPSS View Finder), each company having 5 tables, one for each year (total 60 tables). As it is not possible to append all the tables along with this paper, partial output for only one company (Al-Ahli Commercial Bank) for the 5-year period has been appended for illustration in Appendix – 1 to Appendix – 5.

It was observed from the results that the correlation coefficients are significantly not different from zero values indicating that the changes in daily share prices had been more or less independent and random over 12 month period (12 lag periods) of each of the years.

Correlation coefficients among variables Var01 to Var60 representing the changes in daily share prices for periods ranging from January, 1996 to December, 2000 are found to be not significantly different from zero implying that the percentage changes in share prices are independent and random over the five year period 1996-2000.

3.2 Analysis of Runs Tests based on daily share price changes

The results of Runs Tests conducted for the select 12 firms, year-wise over the 5-year period (1996-2000) at three different levels of significance (1, 5 and 10%). The results for the select 12 companies for the year 2000 are appended (Appendix – 6 and Appendix – 7) for the purpose of illustration.

The close observation of the results of Runs tests of the select sample companies presented in Appendix – 6 and Appendix – 7 may yield the following Table 3 and inferences followed.

Table 3
Select companies showing weak form of Efficient Market Hypothesis
(% in favour)

Year	Level of Significance	1%	5%	10%
1996		9 (75.0%)	8 (66.7%)	8 (66.7%)
1997		11 (91.7%)	10 (83.3%)	10 (83.3%)
1998		10 (83.3%)	10 (83.3%)	10 (83.3%)
1999		12 (100%)	11 (91.7%)	11 (91.7%)
2000		11 (91.7%)	11 (91.7%)	11 (91.7%)

Table 3 reveals that barring year 1996, during the study period i.e. 1997 – 2000, more than 83% of the select companies have shown the weak form of efficient market hypothesis. In the year 1996 also about 67% of the select companies have shown the weak form of efficient market hypothesis, whereas, 75% were in favour of the hypothesis.

3.3 Serial Correlation Tests based on percentage changes in the market returns using BSE Daily Index

The Percentage change in market returns month-wise based on BSE Daily Index over all the 12 months of each of the five years (1996-2000) had been computed using MS-Excel package.

The Correlation coefficients over 12-month period of each of the five years (1996-2000) among the month-wise percentage change in market returns (Var01 to Var60) have been computed. The results for the year 2000 are appended for the purpose of illustration (Appendix – 8)

It was observed from the results that the correlation coefficients are significantly not different from zero values indicating that *the percentage changes in market returns had been more or less independent and random over 12 month period (12 lag periods) of each of the years.*

3.4 Runs Tests based on percentage change in the market return using BSE Daily Index for the years 1996-2000

The results of Runs Tests based on percentage changes in the market returns using BSE Daily Index for the years 1996-2000 are reported in Appendix – 9 and Appendix - 10. It can be inferred from the appendices that the percentage changes in market returns are independent and random at all levels of significance (1, 5 and 10 percent) during the study period 1996 - 2000. *Hence, the Tests support the weak form of Efficient-Market Hypothesis.*

The overall observation of Serial correlation tests and Runs tests over the five- year period (1996-2000) for the select 12 firms confirm the weak form of the Efficient-Market Hypothesis and support the Null Hypothesis.

Thus, it can be inferred that the individual share prices and the BSE Daily Index do fully reflect all the historical information and most of the stocks are priced around its intrinsic values and the price changes are random and independent. *This leads to the conclusion that there is no scope for developing trading strategies and making consistent abnormal gains in the long run.*

This weak form efficiency of the Bahrain Stock Market may be attributed to the well developed trading and regulatory mechanisms and the speed and the transparency with which information is provided to the investors, which have been reviewed objectively in the next what follows:

4 Strategic initiatives of Bahrain Stock Exchange

BSE is striving towards development of Bahrain's financial market in terms of substantial depth and breadth and integration with regional and international financial markets utilizing state-of-the art trading technologies and settlement procedures.

The Bahrain stock market is an emerging market and the current revamping activities are undertaken to transform the market from a well-protected, inwardly- focused one to that of a vibrant regional and international market that offers a wide ranging investment opportunities (depth, and breadth), liquidity, transparency, appropriate regulation to safeguard interests of the investors.

The BSE continues to forge ahead with a development strategy aimed at putting Bahrain on the map of international capital markets. Several initiatives have been launched by BSE to provide an infrastructure that is modern and similar to systems in force in developed capital markets. The ultimate goal of the initiatives is to enable the BSE to play a pivotal role in the national economy by mobilizing private sector saving and attracting foreign investments, through a truly international capital market.

A multi-pronged strategy has been adopted to achieve the above and a brief review is presented below:

- The ground is being carefully laid for the ultimate *opening of the stock market to overseas investors*, which will significantly add to Bahrain's ability to attract foreign

investment, which would develop the financial markets in terms of volatility, market activity, depth and liquidity. Essential elements to such a move are remote and internet trading, both of which are currently being developed. ***The market is currently open to resident expatriates only. Even the ownership of share capital in Bahrain joint stock companies is restricted to non-GCC citizens to the extent of 49% of paid up capital only, whereas, permission was granted for GCC citizens to own up 100% i.e. there is no participation from countries other than GCC.***

- ***Central Depository System (CDS):*** The system, implemented in three phases, is aimed at providing a modern infrastructure for trading to boost share-trading activity on the stock exchange.
- ***The first phase of the CDS*** involved development of appropriate procedures for the settlement of financial transactions. This requires brokers to hold accounts with the Bank of Bahrain and Kuwait (BBK), which, is designated, as the “clearing bank” by the BSE. The moves allow for the speedy settlement of buying and selling transactions, which it is hoped, will boost investors’ confidence.
- ***The second phase of the CDS*** involved the establishment of a depository, which will entail investors to have accounts with the BSE through a designated bank. This will help investors to keep track of their investments in the market.
- ***The third and final phase***, involved setting up of a register of all investments held as shares listed on the BSE.

- One of the most important developments is regarding ***disclosure regulations***. These were initiated to enhance transparency and safety of investors' money with the objective of attracting a large number of small investors. The regulations are aimed to make more information available on the share prices, performance of listed companies and investor activity. ***The approved disclosure standards*** are based on the disclosure standards recommended by IOSCO, and are similar to the standards applied internationally. The standards require the companies already listed as well as future prospective companies to fulfill certain requirements stipulated by the exchange, both during the initial phase and on-going basis. ***These standards have been officially published to improve the transparency.*** BSE joined the International Finance Corporation – IFC Global index (which is expected to enhance the transparency and disclosure) to create investment opportunities to foreign investors (both portfolio and institutional investors).
- The BSE has initiated adequate measures to ***disseminate information***, through its official website as the main channel for improving the standards of transparency and disclosure with regards to its operations. ***Plans are in progress to further upgrade its website to disclose information in the Real-time***, which would further enhance transparency and its reliability towards investors and traders.
- BSE has also initiated measures to ***strengthen the regulations of the exchange*** by introducing new requirements to control insider trading and imposition of certain conditions to control the ownership activities.

- Investors, individuals and companies are required to inform the BSE in writing about their ownership in any listed company if it exceeds five per cent of the company's paid up capital. Also, any person or group intending to acquire more than ten per cent stock in any listed company must inform the Director of BSE of such an intention, prior to placing a buying order on the trading floor. Disclosure of ownership is also required in cases where shares held by members of nuclear family together exceed ten per cent of the paid up capital of any listed company.
- Financial institutions, which are under supervision of Bahrain Monetary Agency (BMA), are required to obtain BMA approval before they acquire a stake exceeding twenty per cent. The BSE also reserves the right to publish the names of shareholders.
- Trading has been *automated* for improving the efficiency of clearance and settlement, which is expected to result in enhanced market activity.
- The institutional arrangements and facilitators only improve the efficiency of the functioning of the stock exchange, but in real sense an efficient market is defined as a market, where there are a large number of rational and profit maximizers actively compete with each other and trying to predict future market values of individual securities, and where important current information is almost freely available to all participants. In other words, in an efficient market, competition exists among the many intelligent participants leading to a situation where, at any point of time, the actual prices of individual securities already reflect the effects of information based on the events that have already occurred and are expected to take place in future, that

is, in an efficient market, at any point of time, the actual price of security will be a good estimate of its intrinsic value.

A close observation of the development and achievements of BSE indicates that BSE requires to continue its present efforts to improve its efficiency, but still a long way to go for attaining a desired place in the global economy and moving towards semi-strong and strong forms of Efficient Market Hypothesis.

5 Strategies for developing GCC Financial Markets

With a view to realize the benefits from the experiences of the developed financial markets, an attempt is also made in this paper to examine the strategies suitable for developing the GCC financial markets and the same have been outlined delineated below:

- ***Integrate the financial markets of GCC countries*** by facilitating simultaneous listing of the securities listed with stock exchanges of GCC countries to overcome the problem of thin trading at each of the stock exchanges. The strategy will also benefit the investors with a wide choice of securities to invest in and enhances market activity in terms of depth and liquidity of the GCC financial markets.
- ***Networking and cooperation among the Stock Exchanges of GCC countries*** to facilitate e-trading/e-broking to provide depth and liquidity to the secondary markets of capital and money markets. This will open a myriad of opportunities in the

financial services sector such as mutual funds, leasing, inter corporate deposits, trading in government and corporate bonds and derivatives, index funds etc.

- ***Create/develop investor awareness*** by educating the mass of small investors regarding the investment opportunities, choices, returns and risks associated with different types of financial securities, how to participate in the financial markets etc. The stock exchanges of GCC countries have a greater role to play in this regard besides better management and regulation of the stock exchanges. The growth and development of financial markets to a large extent depends upon the ability of financial markets to attract small investors.
- Encourage and create a culture of domestic savings by providing a variety of investment opportunities catering to the varied needs of population, particularly the small investors.
- Setting-up of an independent credit rating agency, probably jointly funded by the banks and financial institutions operating in GCC countries, which will award ratings to the securities floated by companies listed on the stock exchanges of GCC countries.

To sum up, the initiatives taken by BSE are in the positive direction of growth and development of Bahrain's financial markets. Further emphasis on some more positive initiatives by BSE coupled with implementation of proposed strategies outlined in the paper would enhance Bahrain's stock market efficiency.

Appendix - 1

Correlation Coefficients for share price changes of Al-Ahli Commercial Bank (1996)

	V01	V02	V03	V04	V05	V06	V07	V08	V09	V10	V011	V12
V01 Pearson	1.000	-.583	.082	.185	-.337	-.043	-.019	.000	-.326	-.246	-.044	-.162
Correlation												
Sig. (2-tailed)	.	.302	.782	.493	.220	.874	.952	1.000	.328	.358	.871	.548
N	16	5	14	16	15	16	12	7	11	16	16	16

Correlation is significant at the 0.05 level (2-tailed).

Appendix - 2

Correlation Coefficients for share price changes of Al-Ahli Commercial Bank (1997)

	V13	V14	V15	V16	V17	V18	V19	V20	V21	V22	V23	V24
V13 Pearson	1.000	-.173	.190	-.134	.145	-.067	-.223	-.484	-.383	-.540	.392	-.094
Correlation												
Sig. (2-tailed)	.	.521	.555	.620	.592	.806	.406	.067	.143	.031	.134	.728
N	16	16	12	16	16	16	16	15	16	16	16	16

Correlation is significant at the 0.05 level (2-tailed).

Appendix - 3

Correlation Coefficients for share price changes of Al-Ahli Commercial Bank (1998)

	V25	V26	V27	V28	V29	V30	V31	V32	V33	V34	V35	V36
V25 Pearson	1.000	.200	.036	.173	.447	.059	.253	.175	.078	.204	.317	.161
Correlation												
Sig. (2-tailed)	.	.513	.894	.521	.168	.828	.344	.517	.773	.448	.231	.552
N	16	13	16	16	11	16	16	16	16	16	16	16

Correlation is significant at the 0.05 level (2-tailed).

Appendix – 4

Correlation Coefficients for share price changes of Al-Ahli Commercial Bank (1999)

	V37	V38	V39	V40	V41	V42	V43	V44	V45	V46	V47	V48
V37 Pearson	1.000	.006	-.245	-.465	.072	-.598	.671	.015	.175	.040	-.066	.166
Correlation												
Sig. (2-tailed)	.	.984	.360	.069	.792	.014	.004	.955	.530	.884	.808	.539
N	16	16	16	16	16	16	16	16	16	16	16	16

Correlation is significant at the 0.05 level (2-tailed).

Appendix - 5

Correlation Coefficients for share price changes of Al-Ahli Commercial Bank (2000)

	V49	V50	V51	V52	V53	V54	V55	V56	V57	V58	V59	V60
V49 Pearson	1.000	-.174	.215	.395	.174	.161	-.163	a	a	a	a	a
Correlation												
Sig. (2-tailed)	.	.463	.378	.116	.462	.522	.577	.	.	.	.	.
N	20	20	19	17	20	18	14	0	0	0	0	0

Correlation is significant at the 0.05 level (2-tailed).

a. Cannot be computed because at least one of the variables is constant.

Appendix - 6

Runs Test for the year 2000
Observed price changes and number of runs

Company	n_1 No. of positive price changes	n_2 No. of negative price changes	R Total no.of runs	μ (mean)	σ (Std. deviation.)
Al-Ahli Commercial Bank	29	46	28	36.57	4.08
Arab insurance Group	35	56	51	44.08	4.49
Bank of Bahrain & Kuwait	32	34	40	33.97	4.03
Bahrain Islamic Bank	26	25	30	26.49	3.53
Bahrain Commercial Facilities	20	26	27	23.61	3.30
Bahrain International Bank	45	64	49	53.84	5.04
Bahrain Middle East Bank	41	83	37	55.89	4.90
Bahrain Telecommunications	40	37	45	39.44	4.35
Investcorp Bank	27	54	44	37	3.97
National Bank of Bahrain	42	43	41	43.49	4.58
National Import & Export	35	40	39	38.33	4.28
United Gulf Bank	40	31	30	35.93	4.11

Appendix - 7

Runs Test for the year 2000
Upper and Lower limits of 'r' at different significance levels

Company	$\alpha = 1\%$	$\alpha = 5\%$	$\alpha = 10\%$
Al-Ahli Commercial Bank	LL:26.10 UL:47.04	LL:28.57 UL:44.57	LL:29.88 UL:43.27
Arab insurance Group	LL:32.50 UL:55.66	LL:35.28 UL:52.88	LL:36.72 UL:51.44
Bank of Bahrain & Kuwait	LL:23.57 UL:44.37	LL:26.07 UL:41.87	LL:27.36 UL:40.58
Bahrain Islamic Bank	LL:17.38 UL:35.60	LL:19.57 UL:33.41	LL:20.78 UL:32.78
Bahrain Commercial Facilities	LL:15.10 UL:32.12	LL:17.14 UL:30.08	LL:18.20 UL:29.02
Bahrain International Bank	LL:40.84 UL:66.84	LL:43.96 UL:63.72	LL:45.57 UL:62.11
Bahrain Middle East Bank	LL:43.25 UL:68.53	LL:46.29 UL:65.49	LL:47.85 UL:63.93
Bahrain Telecommunications	LL:28.22 UL:50.66	LL:30.91 UL:47.97	LL:32.30 UL:46.58
Investcorp Bank	LL:26.76 UL:47.24	LL:29.22 UL:44.78	LL:30.49 UL:43.51
National Bank of Bahrain	LL:31.67 UL:55.31	LL:34.51 UL:52.47	LL:35.98 UL:51.00
National Import & Export	LL:27.29 UL:49.37	LL:29.94 UL:46.72	LL:31.31 UL:38.35
United Gulf Bank	LL:25.35 UL:46.51	LL:27.89 UL:43.97	LL:29.21 UL:42.65

REMARKS: 'r' values at different significance levels are within LL & UL. Price changes are random.

Appendix - 8

Correlation Coefficients for changes in Market Returns (2000)

	V049	V050	V051	V052	V053	V054	V055	V056	V057	V058	V059	V060
V01 Pearson Correlation	1.000	-.194	-.194	.308	.080	.268	.078	.305	.504	.166	.271	.270
Sig.(2-tailed)	.	.591	.591	.228	.816	.401	.743	.336	.056	.554	.294	.372
N	20	10	10	17	11	12	20	12	15	15	12	13

Appendix - 9

Runs Test based on BSE Daily Index

Observed percentage changes in Market Returns and number of runs

Year	n ₁ No. of positive changes in % market returns	n ₂ No. of negative changes in % market returns	R Total no.of runs	μ (mean)	σ (Std. deviation.)
1996	125	102	106	113.33	7.47
1997	146	79	88	103.52	6.79
1998	117	119	97	118.99	7.66
1999	118	106	115	112.68	7.45
2000	94	134	88	112.17	7.31

Appendix - 10

Runs Test based on BSE Daily Index

Upper & Lower limits of 'r' at different significance levels

Year	α = 1%	α = 5%	α = 10%
1996	LL: 94.06 UL: 132.06	LL: 98.69 UL:127.97	LL: 101.08 UL: 125.68
1997	LL: 85.00 UL: 121.04	LL: 90.21 UL:116.82	LL: 92.38 UL: 114.66
1998	LL: 99.23 UL: 138.75	LL:103.98 UL:134.00	LL: 106.43 UL: 131.55
1999	LL: 93.46 UL: 131.90	LL: 98.08 UL:127.28	LL: 100.46 UL:124.90
2000	LL:93.31 UL:131.03	LL: 97.84 UL:126.50	LL: 100.19 UL:124.16

REMARKS: 'r' values at different significance levels are within LL & UL. Price changes are random.

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